

Course Title: Linear Algebra

Course Code: MTH-163

Week 1 — Matrix Algebra

Topics

- Matrix notation
- Matrix addition & multiplication
- Identity & inverse
- Transpose

Applications

Image Processing: 90 Degree Rotation

Week 2 — Determinants

Topics

- Definition
- Determinants by Cofactor Expansion

Week 3 — Inverse of a matrix

- Method for Finding A^{-1}

Applications

Cryptography

Week 4 — Row Operations & Echelon Forms

Topics

- Elementary row operations
- Echelon form

Week 5 — Reduced Echelon Forms

Topics

- Reduced row echelon form (RREF)
- Pivot positions

Application

Solving budget distribution problems

Week 6 — Systems of Linear Equations

Topics

- Matrix form
- Consistent vs inconsistent systems

- Unique vs infinite solutions
- Homogeneous systems

Application

Solving resource allocation problems

Week 7 — LU Decomposition

Topics

- Elimination matrices
- LU factorization

Application

- Faster solving of repeated systems
- Basic numerical computation concept

Week 8 — Vectors & Linear Independence

Topics

- Vectors in
- Linear combinations
- Linear independence
- Span

Application

Feature representation in machine learning

Week 9 — Basis and Dimension

Topics

- Basis
- Dimension
- Column space
- Null space

Application

Dimensionality Reduction

Week 10 — Linear Transformations

Topics

- Definition
- Matrix representation

Week 11— Null Space and Column Space

- Kernel and range

Applications

Image rotation and scaling

Week 12 — Eigenvalues & Eigenvectors

Topics

- Characteristic equation

Applications

Application to Dynamical Systems

Week 13 — Diagonalization

Topics

- Diagonalization
- Matrix powers

Week 14 — Dot Product & Orthogonality

Topics

- Dot product
- Orthogonal vectors
- Orthogonal projections

Applications

Data Projection

Week 15 — Least Squares

Topics

- Least squares method

Applications

Basic regression idea

Week 16 — Gram–Schmidt

Topics

- Orthonormal basis
- Gram–Schmidt